Last week

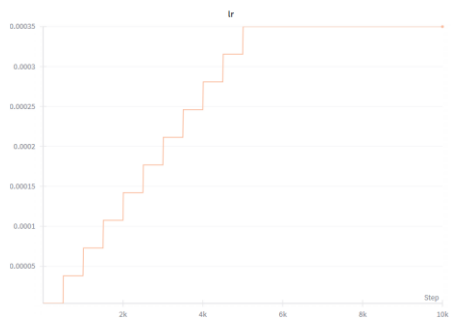
1. [**Training**]: Fix the learning rate scheduler such that it anneals over time instead of increasing; redo all the experiments
2. [**Experiment**]: Combine all of the improvements in this week into one model, going step-by-step to understand the contribution of each modification
   - Do cross-validation on the evaluation/test set to rule out random seeding behaviors
3. [**Model**]: ~~Fix the LogSumExp activation function / approximation and re-run the test~~
4. [**Theory/Experiment**]: Brainstorm a better approximation of the Rander's drift vector compared to the symmetric trapezoidal approximation
5. [**Theory/Experiment**]: Consider the improvements in reducing the drift vector dimension as a potential scaling problem (multiply by N/S) – substantiate this argument and test it out

Put on Hold

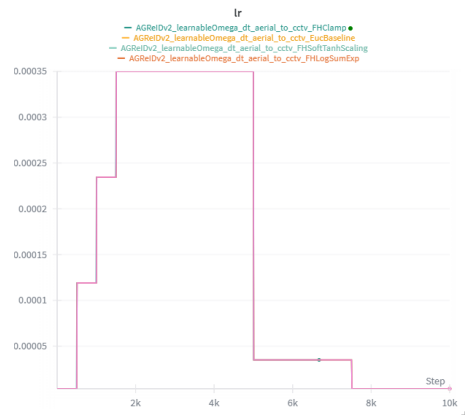
1. [**Loss**]: Consider implementing different repulsive forces to enforce domain generalization (other than uniformity and domain loss). *Idea*: Something similar to the Finsler stress loss
2. [**Sanity Test**]: Devise a small experiment to test the logic behind cross-res asymmetry and camera-viewpoint bias (e.g., small dataset + simple metric).

[Training] Modified Learning Rate Scheduler

Before



Now



- Before we used an unscaled learning rate schedule where the default step-decay milestones ($M = [30, 50]$) fall outside the truncated 20-epoch limit, resulting in an absence of learning rate decay. I've proportionally scaled both the linear warmup phase and the milestone decay steps to strictly fit within the 10k step horizon ($E = 20$), restoring the required gradient attenuation.

- To mathematically isolate the contributions of the proposed Finsler manifold formulation from hyperparameter-induced variance, the canonical BAU Euclidean ResNet baseline is optimized and evaluated under this exact identical compressed 20-epoch schedule.

- Compressing the optimization budget is theoretically suboptimal for deep metric learning manifolds. Truncating the training forces the network to terminate during a high-variance transient phase, limiting the capacity for fine-grained feature disentanglement and preventing the embeddings from reaching a stable, uniform distribution.

- This truncated schedule is a strictly temporary heuristic. It deliberately sacrifices asymptotic convergence bounds to maximize experimental

throughput, enabling rapid iteration across architectural variants and Randers metric parameterizations during the preliminary search phase.

- **Canonical Scaling Protocol:** Upon empirical validation of the optimal architectural formulation, the training protocol will definitively revert to the canonical $E = 60$ regime. The final evaluation will utilize the complete multi-domain composite dataset (Market1501, MSMT17, CUHK-SYSU, AGReID) to ensure robust gradient convergence, minimize source-domain overfitting, and establish rigorous state-of-the-art validation.

EXTRAS

In the context of deep neural network optimization and metric learning, the **transient phase** (often referred to as the early-training transient regime) denotes the initial and intermediate stages of gradient descent where the learning rate remains at its peak or warmup maxima. Mathematically and geometrically, this phase is characterized by:

- **High-Variance Gradient Updates:** The large step sizes prevent the optimizer from settling into sharp local minima. Instead, the parameters traverse the macro-structure of the loss landscape.
- **Coarse-Grained Embedding Displacements:** In the latent metric space (e.g., the hypersphere or Finsler manifold), the representations undergo massive structural shifts. The repulsive forces of the triplet or contrastive loss push dissimilar instances apart, but the high kinetic energy of the optimizer prevents fine-grained, exact disentanglement.
- **Absence of Asymptotic Uniformity:** To achieve a theoretically optimal representation, the feature vectors must be distributed uniformly across the latent manifold (maximizing entropy and minimizing hyperspherical energy). During the transient phase, the continuous injection of optimization noise—driven by the high learning rate and mini-batch stochasticity—causes the embeddings to constantly oscillate.

Only after the learning rate decays (entering the *stationary* or *asymptotic phase*) can the network execute micro-adjustments, allowing the contrastive gradients to strictly enforce fine-grained feature disentanglement and uniform packing without overshooting. Truncating the training before this decay forces the model to output representations that are mathematically unstable and strictly sub-optimal for open-set retrieval.

Primary Citation for DML Uniformity and Convergence:

Wang, T., & Isola, P. (2020). Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. *In International Conference on Machine Learning (ICML)*.

- *Relevance:* Wang and Isola mathematically formalize that optimizing the contrastive objective requires asymptotic convergence to minimize the hyperspherical energy (the "uniformity" metric). Truncating optimization prematurely prevents the embeddings from satisfying this uniformity bound, leaving the latent space highly localized and poorly generalized.

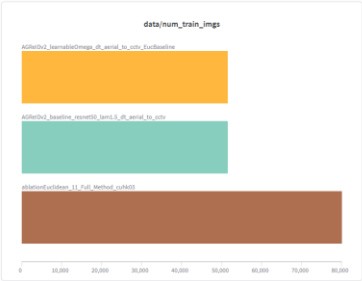
Primary Citation for Transient Phase Dynamics and DML Generalization:
Roth, K., Milbich, T., Sinha, S., Gupta, P., Ommer, B., & Cohen, J. P. (2020). Revisiting Training Strategies and Generalization Performance in Deep Metric Learning. *In International Conference on Machine Learning (ICML)*.

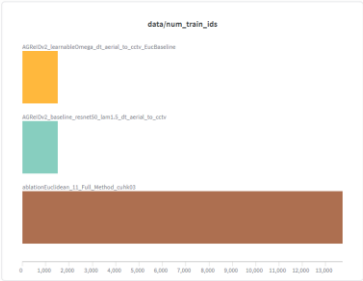
- *Relevance:* Roth et al. explicitly investigate the impact of training durations, batch scheduling, and learning rate decays in Deep Metric Learning. They demonstrate that DML models evaluated during early or truncated epochs exhibit high-variance representations that fail to capture diverse, fine-grained visual characteristics. The paper strictly proves that stable, disentangled feature representations strictly require completing the full optimization schedule, validating your critique of the 20-epoch truncation.

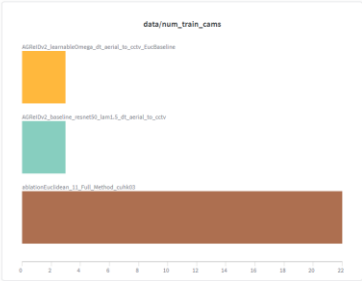
Reference for the Terminology "Transient Phase":
Lewkowycz, A., Bahri, Y., Roelofs, R., Alonso, E., & Sharma, V. (2020). The large learning rate phase of deep learning: the catapult mechanism. *arXiv preprint arXiv:2003.02218*.

- *Relevance:* This work formalizes the "transient phase" of optimization, proving that while high initial learning rates are necessary to escape sharp, poor-generalizing basins, the network must subsequently transition out of this high-variance transient phase via step-decay to achieve optimal feature separation.

[Training] Why only 20 epochs?







[Test] Why Re-ID does not support conventional cross-validation

- **The Open-Set Paradigm**: Train and test identities must remain strictly disjoint to evaluate true generalization, not memorization.
- **Data Leakage**: Standard k-fold cross-validation randomly splits images, leaking training identities into validation folds and violating open-set rules [1].
- **Complex Camera Topology**: ReID requires matching a *Query* (1 camera) against a *Gallery* (different cameras). Dynamically generating valid cross-camera splits per fold is error-prone.
- **Standardization for Benchmarking**: Metrics (mAP, Rank-1) are extremely sensitive to the number of gallery distractors. A single, fixed split is mandatory for fair model comparison [2].

***Alternatively*:** We can train the final models with different random seeds and average out the metrics (like we did before)

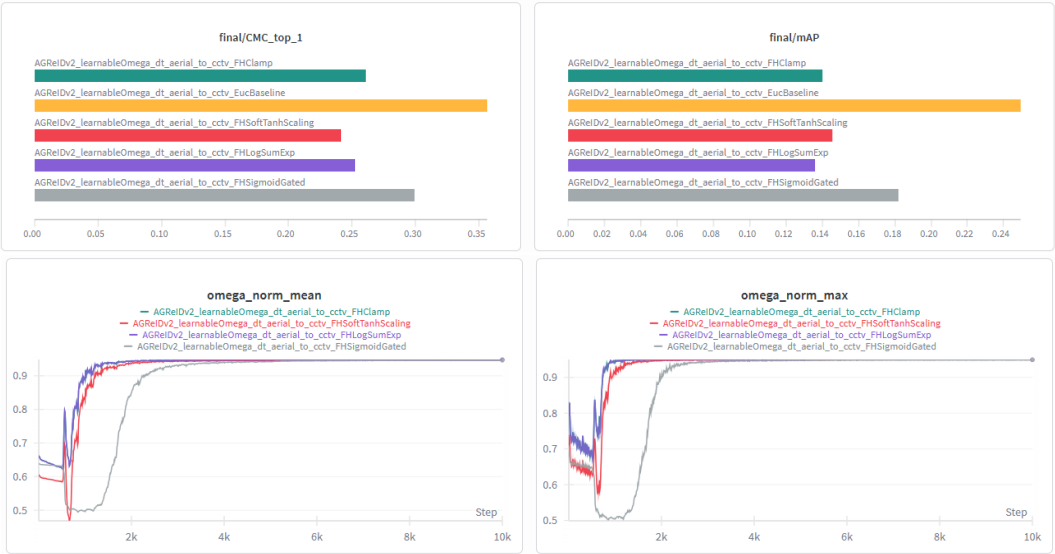
[1] Ye et al., "Deep Learning for Person Re-identification: A Survey and Outlook" (IEEE TPAMI 2021).
[2] Zheng et al., "Scalable Person Re-identification: A Benchmark" (ICCV 2015).

- Applying cross validation to Person Re-Identification breaks the fundamental rules of the task.
- ReID is about metric learning—teaching a model *how* to compare people, not *who* they are. If we use standard cross-validation, the same person appears in both training and testing. The model just memorizes their specific shirt or appearance, which artificially inflates the score and ruins our ability to test on unseen people.
- 'Why not just split the folds by identity?' To prove a ReID model works, we must guarantee we are matching a person from Camera A to Camera B. Creating perfectly balanced, unbiased Query and Gallery splits across multiple folds that respect this cross-camera rule is mathematically messy and introduces severe variance.
- Furthermore, the primary metrics—Rank-1 accuracy and Mean Average Precision (mAP)—are highly dependent on the gallery size. Finding one person among 1,000 distractors is much easier than finding them among 20,000. If folds have different gallery distributions, the scores become meaningless.

Conclusion: Because of this, since the landmark Market-1501 paper in 2015, the community universally agreed to use exactly one fixed, heavily standardized data

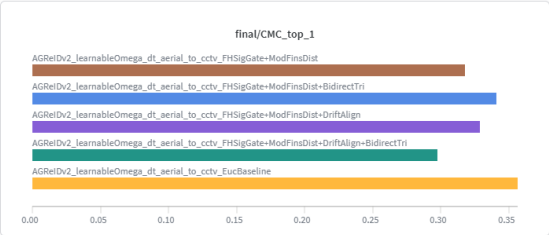
split per dataset. This ensures absolute fairness when benchmarking models like we see in codebases like BAU and FastReID.

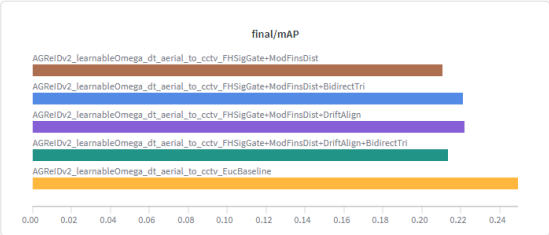
[Model] Improving the Finsler Drift Head (Pt. 2)

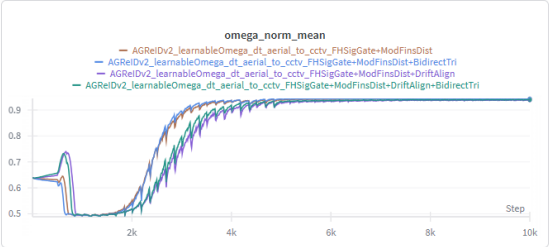


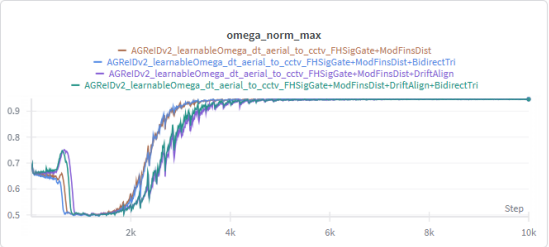
Moving forward, we are benchmarking models that have the sigmoid gate implemented in the Finsler head

[Model+Loss] Combining all of the Improvements





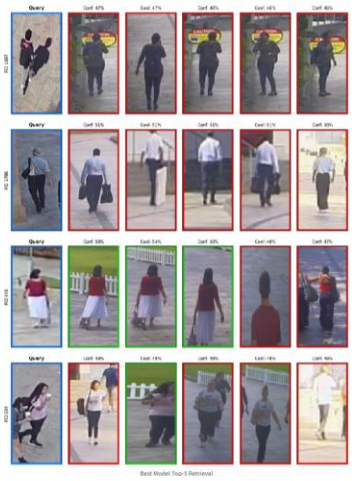




[Model+Loss] Combining all of the Improvements

ModFistDist+DriftAlign+BidirectTri

EucBaseline



[Model+Loss] Combining all of the Improvements

ModFistDist

ModFistDist + BidirectTri

ModFistDist + DriftAlign

[Distance] Improve the geodesic approximation in Finsler distance computation

We make two key simplifications:

- **Path Assumption**: Taking straight Euclidean line segment as the path between x and y
- **Field Assumption**: The drift vector field w changes linearly along this path

1 $$\gamma(t) = \mathbf{x}_{id} + t(\mathbf{y}_{id} - \mathbf{x}_{id}) \qquad \dot{\gamma}(t) = \mathbf{y}_{id} - \mathbf{x}_{id}$$

2 $$\mathcal{A}(\mathbf{x}, \mathbf{y}) = \int_0^1 \langle \omega(\gamma(t)), \dot{\gamma}(t) \rangle dt$$

3 $$\omega(\gamma(t)) \approx (1-t)\mathbf{x}_\omega + t\mathbf{y}_\omega \qquad \mathcal{A}(\mathbf{x}, \mathbf{y}) \approx \int_0^1 \langle (1-t)\mathbf{x}_\omega + t\mathbf{y}_\omega, \mathbf{y}_{id} - \mathbf{x}_{id} \rangle dt$$

$$\int_0^1 (1-t)dt = \frac{1}{2}, \quad \int_0^1 t dt = \frac{1}{2}$$

4 $$\mathcal{A}(\mathbf{x}, \mathbf{y}) = \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id} \rangle$$

$$\mathcal{A}(\mathbf{x}, \mathbf{y}) = \frac{1}{2}\left[(\mathbf{x}_\omega \cdot \mathbf{y}_{id}) - (\mathbf{x}_\omega \cdot \mathbf{x}_{id}) + (\mathbf{y}_\omega \cdot \mathbf{y}_{id}) - (\mathbf{y}_\omega \cdot \mathbf{x}_{id})\right]$$

Symmetrical Trapezoidal Approximation

1. **Define the Path**
   - We define the path $\gamma(t)$ in the identity feature space from x_{id} to y_{id} , parameterized by $t \in [0, 1]$.
   - The velocity vector (tangent to the path) is constant:

2. **Define the Integral**
   The asymmetric component $A(x, y)$ of the Randers metric is defined as the path integral of the drift one-form ω acting on the velocity vector.

3. **Linear Interpolation (Trapezoidal Approximation)**
   By assuming the drift vectors change linearly, we approximate $\omega(\gamma(t))$ as a weighted sum of the drift at start (x_ω) and end (y_ω).
   - Substituting this back into the integral
   - Since the velocity term is constant with respect to t, we integrate the scalar coefficients

4. **Step 4: The Closed Form Solution**
   - This simplifies the integral to the dot product of the **average drift vector** and the displacement vector.
   - Finally, we expand this dot product into the four specific terms

5. X and y have the same dimensions

Randers distance of the arc circle?

[Distance] Improve the geodesic approximation in Finsler distance computation

We make two key simplifications:

- **Path Assumption**: The geodesic is approximated as a straight Euclidean line segment
- **Field Assumption**: The drift vector field w changes linearly along this path

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id} \rangle$$

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \frac{1}{2}\left[(\mathbf{x}_\omega \cdot \mathbf{y}_{id}) - (\mathbf{x}_\omega \cdot \mathbf{x}_{id}) + (\mathbf{y}_\omega \cdot \mathbf{y}_{id}) - (\mathbf{y}_\omega \cdot \mathbf{x}_{id})\right]$$

[Model]: Refining the Rander's Drift: Spherical Trapezoidal Approximation

Recall: The *Symmetric Trapezoidal Approximation* assumed a linear Euclidean path between embeddings x and y.

$$\gamma(t) = x + t(y - x)$$

The Flaw: L_2-normalized identity features strictly reside on a hypersphere (S^{d-1}).

- Thus, $\| x \|_2 = 1$ and $\| y \|_2 = 1$.
- A straight Euclidean line cuts through the *interior* of the sphere, leaving the valid data manifold.

Defining the Geodesic Path: Integrate the drift vector ω along the true geodesic (a great circle arc).

- The angle between them: $C = \cos(\theta) = \langle x, y \rangle$

Deriving the Spherical Tangents:
To integrate the drift vector ω along this arc, we need the path's tangent vectors. We find these by projecting the destination point onto the tangent space of the origin point ($T_x S^{d-1}$).

- *Tangent at x pointing towards y:* $v_x = y - proj_x(y) = y - \langle x, y \rangle x = y - Cx$
- *Tangent at y arriving from x:* $v_y = Cy - x$

The Final Asymmetry Term:

$$\mathcal{A}(\mathbf{x}, \mathbf{y}) \approx \frac{1}{2}\left[\langle \mathbf{x}_\omega, \mathbf{y}_{\text{id}} - C\mathbf{x}_{\text{id}} \rangle + \langle \mathbf{y}_\omega, C\mathbf{y}_{\text{id}} - \mathbf{x}_{\text{id}} \rangle\right]$$

Connecting to previous slides:

- Previousy, we introduced the Symmetric Trapezoidal Approximation to compute the Randers distance.
- To make the integral tractable, we assumed the shortest path between query x and gallery y was a straight Euclidean line.

Identifying the theoretical flaw:

- However, because our network uses L_2 normalization on the identity features, our valid embedding space is not flat—it is a hypersphere S^{d-1}
- By integrating along a straight line $\gamma(t) = x + t(y - x)$,the midpoint of our path has a norm less than 1 ($\| \frac{x+y}{2} \| < 1$).
- We are effectively evaluating the drift vector ω in the empty interior of the sphere, where the network has never been trained to map features.

The Geodesic Fix:

- To fix this, we must integrate along the true shortest path on a hypersphere: a great circle arc.
- The angle between them is given by their cosine similarity: $C = \cos(\theta) = \langle x, y \rangle$

Deriving Spherical Tangents

- Instead of the constant Euclidean velocity $y - x$, we calculate the unnormalized tangent vectors along the spherical arc.
- At point x, the tangent pointing towards y is $v_x = y - \cos(\theta)x$. We subtract the component of y that is parallel to x.
- At point y, the tangent arriving from x is $v_y = y\cos(\theta) - x$

The Final Asymmetry Term:

- Substituting these spherical tangents into the trapezoidal rule yields our new, manifold-aware asymmetry penalty

Why this improves performance:

- By projecting the path tangents onto the hypersphere, we ensure that the asymmetric penalty is calculated strictly using valid, on-manifold feature representations.
- This prevents the distance metric from being distorted by off-manifold 'shortcuts', leading to a much more accurate estimation of the information degradation between the aerial and ground views.

Extras

3. How it changes the Integral

Because the new velocity $\dot{\gamma}(t)$ is not constant, we cannot easily do what Step 3 (pulling $y_{id} - x_{id}$ out of the integral).

Instead, we approximate the integral $\int_0^1 \langle \omega'(t)\dot{\gamma}(t)\rangle dt$ by averaging the dot product at the start and the end (this is where the Trapezoidal Rule explicitly comes in):

At $t = 0$**:** $\langle \omega'(0)\dot{\gamma}(0)\rangle = \langle x_{\omega'}\, y_{id} - Cx_{id}\rangle$

At $t = 1$**:** $\langle \omega'(1)\dot{\gamma}(1)\rangle = \langle y_{\omega'}\, Cy_{id} - x_{id}\rangle$

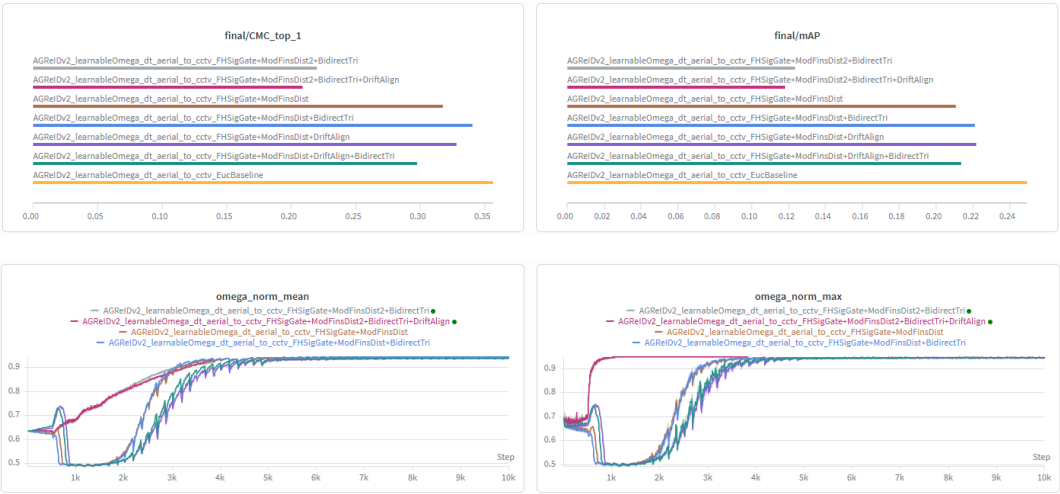
Theoretical Substantiation (if asked):

On Hyperspherical embeddings: "As shown by Wang & Isola (ICML 2020) in their work on alignment and uniformity, L_2 normalized representation learning strictly confines meaningful semantic interpolation to the surface of the hypersphere."

On Geodesics: "Standard Riemannian geometry (e.g., Lee's *Introduction to Riemannian Manifolds*) dictates that the tangent space T_xS^{d-1} is the orthogonal complement to x. The projection of y onto T_xS^{d-1} is exactly $y - \langle x, y\rangle x$,which gives us our corrected velocity vectors."

Compute the geodesic path
Slerp explicit formula that gives you the location of the trajectory – explicit formula for the trajectory
If the trajectory is on a sphere, the omegas are supposed to be tangent to the hypersphere -> learn omega on the tangent plane? How do we enforce that
Identity – XX^t is on the tangent of the sphere, where x is normalized

[Model]: Refining the Rander's Drift: Spherical Trapezoidal Approximation



Observation: We see a distinct drop in mAP and Rank-1 accuracy, and the latent space clusters are noticeably less aligned than our initial Euclidean approximation.

Analysis (The "Vanishing Asymmetry" Problem): While mathematically correct, the spherical projection introduces a dynamic scaling factor via the cosine similarity (C).

- When features belong to the same identity and get close ($C \to 1$), the tangents approach $y - x$ and $y - x$, causing the asymmetry to vanish.
- This prevents the drift vector from applying fine-grained corrections within tight clusters, leading to the observed cluster misalignment.

Analysis (Gradient Entanglement):

- In the Euclidean approximation, the gradient of the asymmetry term with respect to x was relatively straightforward.
- Now, the asymmetry term is heavily multiplied by $C = \langle x, y \rangle$ .This means the network can minimize the asymmetric triplet loss by simply manipulating the cosine similarity (pushing identity features apart) rather than learning a meaningful drift vector ω.

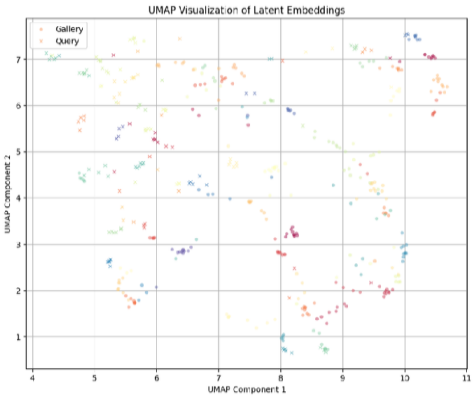
Observation: The ω (drift) norm and mean curves take a completely different shape compared to previous runs.
Analysis (Restricted Drift Growth): Because the self-projection terms ($\langle \omega_{x'} x \rangle$) are now scaled by C, the effective magnitude of the drift vector is suppressed when comparing distant domains ($C \approx 0$).

- The network struggles to establish a consistent magnitude for ω because its impact on the total distance is constantly fluctuating based on the angle between the identity features. This leads to erratic or stunted norm curves as the drift head fails to find a stable optimization trajectory.

[Model]: Refining the Rander's Drift: Spherical Trapezoidal Approximation

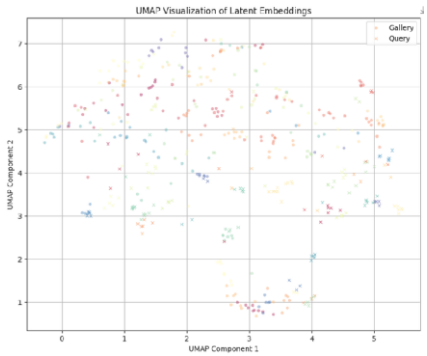
ModFistDist2+BidirectTri

ModFistDist+BidirectTri

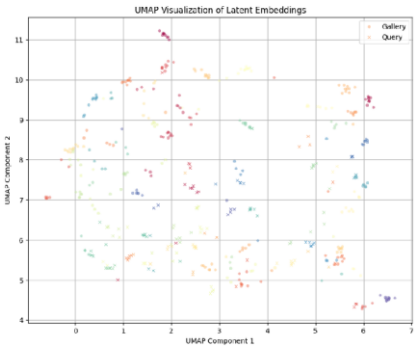


[Model]: Refining the Rander's Drift: Spherical Trapezoidal Approximation

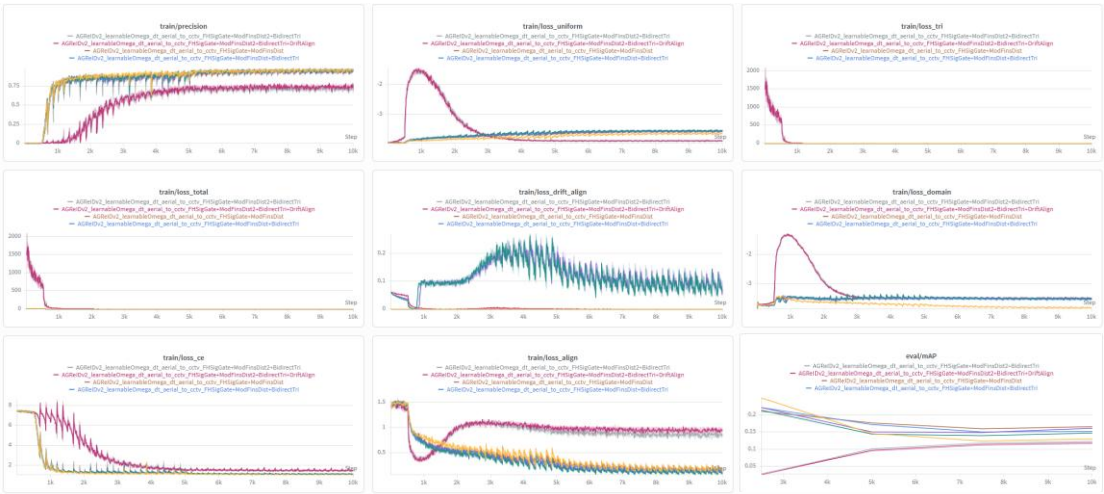
ModFistDist2+BidirectTri+DriftAlign



ModFistDist+BidirectTri+DriftAlign



[Model]: Refining the Rander's Drift: Spherical Trapezoidal Approximation



Triplet & Alignment Loss Dynamics

- **Observation:** The triplet loss starts exceptionally high, and the alignment loss (which was previously stable) begins to increase after an initial period.
- **Analysis (The Identity-Drift Conflict):** The high initial triplet loss shows that the C-scaled asymmetry severely disrupts the initial relative distances of the batch.
- **Analysis (Why Alignment Increases):** The rising alignment loss is the smoking gun.
    - Recall that our alignment loss operates *only* on the identity features.
    - Because the new triplet loss calculates asymmetry using C (the dot product of identity features), the triplet loss backpropagates strong gradients directly into the identity backbone to satisfy the margin.
    - It is actively pushing positive identity features apart to manipulate C, which directly fights and worsens the alignment loss.

Uniform/Domain Loss Peaks & Poor Precision

- **Observation:** Both uniform and domain losses show a sudden peak before settling, and overall training precision remains poor.
- **Analysis (Latent Space Reorganization):** The peaks in the uniform and

domain losses indicate a sudden, violent reorganization of the latent space. As the network tries to push domains apart, the non-linear nature of the spherical tangents causes the embeddings to "snap" into a suboptimal configuration.

- **Analysis (Precision Collapse):** The poor training precision confirms that the model is failing to confidently classify identities. The theoretical strictness of the geodesic path has over-constrained the network.
- By forcing the drift to perfectly adhere to the hypersphere's surface, we removed the "slack" the network previously used in the Euclidean approximation to easily separate identities, resulting in a mathematically pure but empirically weaker representation.

[Sweep] Drift Vector Dimension Sweep

Previously:

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\text{id}} - \mathbf{y}_{\text{id}}\|_2 + \sum_{k=1}^{S} \mathbf{x}_{\omega}^{(k)} (\mathbf{y}_{\text{id}}^{(k)} - \mathbf{x}_{\text{id}}^{(k)})$$

$$S = \min(\dim(\mathbf{x}_{\text{id}}), \dim(\mathbf{x}_{\omega}))$$

Now:

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\text{id}} - \mathbf{y}_{\text{id}}\|_2 + \frac{N}{2S} \sum_{k=1}^{S} \left(\mathbf{x}_{\omega}^{(k)} + \mathbf{y}_{\omega}^{(k)}\right) \left(\mathbf{y}_{\text{id}}^{(k)} - \mathbf{x}_{\text{id}}^{(k)}\right)$$

$$N = \dim(\mathbf{x}_{\text{id}})$$

$$S = \min(\dim(\mathbf{x}_{\text{id}}), \dim(\mathbf{x}_{\omega}))$$

- **Addressing previous feedback:** Investigated why reducing the drift dimension (S) from 2048 down to 16 seemed to artificially improve performance.
- **The Scaling Artifact:** Thomas' intuition was correct. The asymmetry term relies on a dot product. Summing over 16 elements instead of 2048 naturally produces a much smaller value, causing the distance function to degenerate back to the (better-performing) Euclidean baseline.

Why couldn't the network just learn larger weights to compensate? In a standard network, the optimizer would just scale up the ω vector. Here, the model is mathematically handcuffed by two constraints:

- **The Randers Constraint:** To keep the metric strictly positive definite, the L2-norm of the drift vector ($\| \omega \|$) is strictly capped at 0.95 by our FinslerDriftHead.
- **Identity Slicing:** The identity vector (y_{id} (is L2-normalized over all 2048 dimensions ($\| y_{id} \| = 1$).

Calculating the 11x Reduction:

- By the Cauchy-Schwarz inequality, the absolute maximum magnitude of our

asymmetry dot product is $\| \omega \| \times \| y_{id_slice} \|$.

- **At $S = 2048$:** We use the full identity vector. The max magnitude is $0.95 \times 1.0 = 0.95$.
- **At $S = 16$:** We only slice the first 16 dimensions of the normalized identity vector. Assuming features are roughly evenly distributed, the expected L2-norm of this 16-dim slice is $\sqrt{16/2048} \approx 0.088$.
- Therefore, the max magnitude at $S = 16$ is $0.95 \times 0.088 \approx 0.084$.
- Comparing the two maximums ($0.95 \div 0.084$), the asymmetry term at $S = 16$ is forced to be roughly **11 times smaller**, even when the model pushes the drift vector to its absolute limit.

The New Setup:

- To test if the previous improvements were purely this scaling artifact, we introduced the N/S scaling factor (where $N = 2048$).
- This artificially restores the magnitude of the lower-dimensional dot products to match the 2048-dim baseline.
- If the $S = 16$ model with this scaling performs poorly again, it proves the improvement was just an artifact of vanishing asymmetry, not genuinely better generalization.

EXTRA

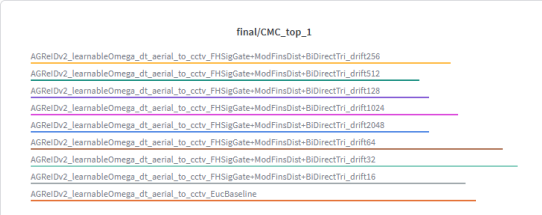
The Cauchy-Schwarz inequality is a fundamental principle in linear algebra that establishes an upper bound on the dot product (inner product) of two vectors based on their individual lengths (magnitudes).

The Formula

For any two vectors u and v, the inequality states:

$$| \langle u, v \rangle | \leq \| u \| \| v \|$$

[Sweep] Drift Vector Dimension Sweep



Rerun the experiment
Spherical interpolation with and without truncation

Hypothesis Confirmed: The introduction of the N/S scaling factor successfully decoupled the drift vector's dimensionality from the magnitude of the asymmetry term.

Flattened Performance Curve:

- Looking at the final/mAP and final/CMC_top_1 bar charts, the massive performance gap between lower dimensions ($S = 16$) and higher dimensions ($S = 2048$) has largely vanished.
- Previously, $S = 16$ artificially approached the Euclidean baseline. Now, with the asymmetry magnitude mathematically restored, $S = 16$ performs similarly to the rest of the pack (hovering around ~22-25% mAP).

The "Improvement" was an Illusion:

- Because lower dimensions no longer dominate when scaled fairly, we can conclude they were *not* learning a more generalizable representation.

- The previous success of $S = 16$ was purely a mathematical artifact: the asymmetry term became so small it effectively turned the model back into a standard Euclidean network.

Network Behavior (Omega Norms):

- The omega_norm_max and omega_norm_mean charts reveal that the network rapidly pushes the drift vector (ω) to its absolute ceiling (the 0.95 max_norm constraint).
- This indicates the optimizer is actively trying to maximize the drift magnitude to satisfy the loss functions, but is hard-capped by the strict Randers metric constraints.

Strategic Takeaway:

Next Steps: Instead of tuning the dimension size, we need to rethink *how* the asymmetry is applied or regularized. The current constrained drift might be too disruptive to the underlying identity manifold.

Regularization: -log(1-w); heuristic regularizers to prevent divergence of the omega norm

Or 1/(1-|w|) ; inside can be squared, outside too

Today

1. [**Literature**]: Consult the SOTA for AGReIDv2 and the protocols from BAU, and benchmark the current top Finsler model against them. Decide or not to pursue publication given results.
2. [**Training**]: Train the baseline and best model configuration with full 60 epochs
3. [**Training**]: Train the best model configuration on one of BAU's training dataset
4. [**Distance**]: Use slerp interpolation to integrate on a spherical geodesic for option 2
   - Consider methods to learn omega on the tangent plane and enforce that
   - Example: Identity – XX^t is on the tangent of the sphere, where x is normalized
5. [**Model**]: Enforcing tangent plane orthogonality
6. [**Model**]: Implement heuristic regularization terms to prevent divergence of the omega norm
7. [**Theory**]: Explain why drift align and bidirectional triplet are incompatible regularization terms together
8. [**Theory**]: Explain why having a 32-dimensional drift vector is optimal, compared to 64 and 16; and why the drift vector does not diverge to the max in this specific scenario

[**Sanity Test**]: Devise a small experiment to test the logic behind cross-res asymmetry and camera-viewpoint bias (e.g., small dataset + simple metric).

[**Literature**]: Current SOTA for AGReIDv2 and BAU

Found that the premise of integrating an asymmetric Finsler space into a domain-generalizable person Re-ID space is theoretically challenging

- Person Re-ID is fundamentally an identity retrieval task, which necessitates a equivalence relation governed by metric symmetry
- Introducing distance asymmetry inherently violates identity equivariance, as the retrieval ranking becomes strictly dependent on the query-gallery orientation.

Train: Market1501 + MSMT17 + CUHK-SYSU → Test: CUHK03

- **CLIP-ReID with Depth-First Graph Sampling:** 50.4% mAP, 51.1% Rank-1
  - Architecture: ViT-B/16 with text-token prompt learning and graph-based hard negative mining.
- **BAU:** 42.8% mAP, 43.9% Rank-1

AG-ReIDv2: Aerial → Camera

- **V2E (Baseline):** 80.72% mAP, 88.77% Rank-1.
  - Architecture: Three-stream ResNet/ViT hybrid with explainable elevated-view attention.
- **SD-ReID**: 80.61% mAP, 87.04% Rank-1.
  - Generatesview-specific features via stable diffusion to bridge extreme perspective gap

The equality operator over a set of identities constitutes a mathematical equivalence relation because it inherently satisfies three axioms: reflexivity, symmetry ,and transitivity

Because the underlying identity relation is symmetric ($y_i = y_j$ is identical to $y_j = y_i$,(the continuous distance function used to measure their similarity must also be symmetric:

$$d(x_i, x_j) = d(x_j, x_i)$$

.If a distance function is asymmetric (as in a Finsler/Randers metric where $F(x_i, x_j) \neq F(x_j, x_i)$,(it violates the geometry of the equivalence class. It implies

x_i

is closer to

x_j

than

x_j

is to

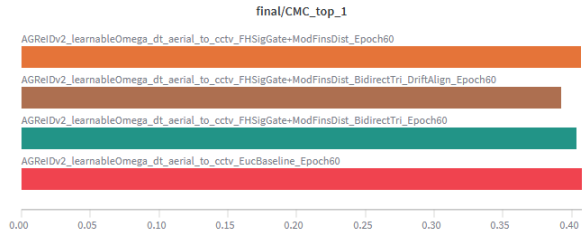
x_i

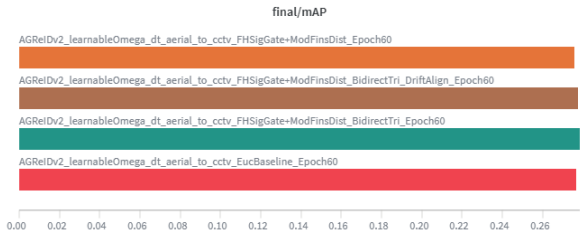
,which contradicts the binary, bidirectional nature of identity matching.

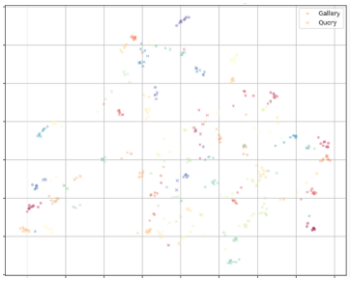
- **Incompatible Objectives:** BAU is a DG-ReID framework. Its primary objective (alignment and uniformity) is to suppress domain-specific stylistic variations (e.g., lighting, resolution) to extract a universal, domain-agnostic identity representation.
- **Deterministic Geometry vs. Style Shift:** AG-ReID.v2 exhibits severe deterministic geometric transformations. This is not a "style shift." SOTA approaches on this dataset (like SD-ReID and V2E) explicitly retain and model the view-specific structure rather than projecting it into a uniform hypersphere.
- **Topological Conflict:** Grafting an asymmetric Finslerian drift vector ω onto BAU and evaluating it on AG-ReID.v2 means your loss function is simultaneously trying to collapse domains into a uniform hypersphere (via BAU's uniformity loss) while trying to force directed, asymmetric distances based on structural elevation (via the Randers metric). These gradients will destructively interfere.

- If your objective is to evaluate a Finslerian manifold for asymmetric cross-view matching, you must benchmark against AG-ReID.v2 SOTAs using view-aware architectures, completely abandoning BAU and its uniformity constraints.
- If your objective is domain generalization, you must evaluate on the M+MS+CS → C3 split using standard VFM baselines like CLIP-ReID, and justify mathematically how an asymmetric distance metric preserves cross-domain identity equivariance.

Training on AGReIDv2 for 60 Epochs



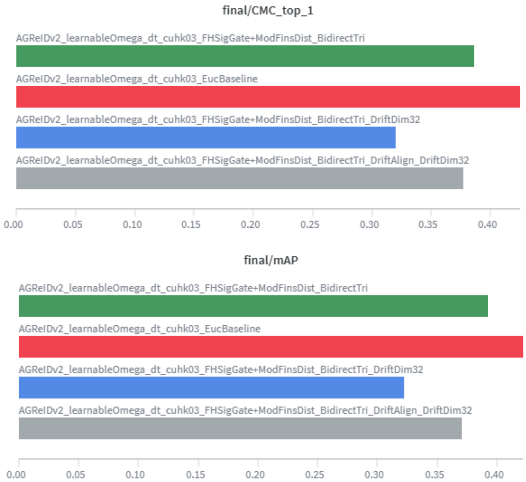


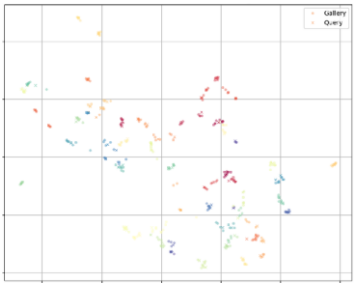


Green

Baseline

Training on M+MS+CS → C3 (Protocol 2) for 60 Epochs





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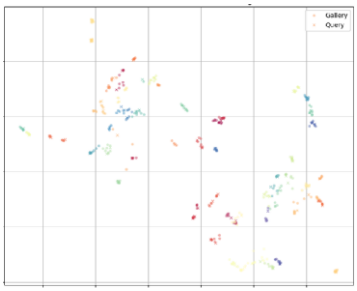


Baseline

[Distance 1] Using Slerp Interpolation

- **Spherical geodesic as path**: Taking spherical geodesic as the path between x and y
- **Field Assumption**: The drift vector field w changes linearly along this path

1 $$\mathcal{A}(\mathbf{x}, \mathbf{y}) = \int_0^1 \langle \omega(\gamma(t)), \dot{\gamma}(t) \rangle dt$$

2 $$\theta = \arccos(\langle x_{id}, y_{id} \rangle) \quad | \quad \gamma(t) = \frac{\sin((1-t)\theta)}{\sin(\theta)} x_{id} + \frac{\sin(t\theta)}{\sin(\theta)} y_{id}\,, \quad t \in [0, 1]$$

3 $$\dot{\gamma}(t) = A(t) x_{id} + B(t) y_{id}$$

where, $$A(t) = -\frac{\theta}{\sin(\theta)} \cos((1-t)\theta) \quad , \quad B(t) = \frac{\theta}{\sin(\theta)} \cos(t\theta)$$

4 $$\omega(t) \approx (1-t) x_\omega + t y_\omega$$
$$\langle (1-t) x_\omega + t y_\omega, A(t) x_{id} + B(t) y_{id} \rangle$$
$$(1-t) A(t) \langle x_\omega, x_{id} \rangle + (1-t) B(t) \langle x_\omega, y_{id} \rangle + t A(t) \langle y_\omega, x_{id} \rangle + t B(t) \langle y_\omega, y_{id} \rangle$$

1. In Randers spaces, the distance from x to y is defined as the standard Riemannian distance plus a continuous line integral of a vector field W over the connecting curve γ
2. Given the great-circle angular distance between two identity features, we can define the spherical trajectory that is parameterized via SLERP
3. To evaluate the integral, we need the tangent velocity vector. Taking the direct analytical derivative of the SLERP formulation with respect to t yields:
4. The continuous ambient drift vector $W(\gamma(t))$ must be evaluated at every step. We approximate the intermediate drift field via linear interpolation of the anchor drift representations W_x and W_y
    1. Because these spatial interactions are decoupled from the time parameter t, they are pre-computed outside the integration loop via $O(1)$ batch matrix multiplications, drastically improving computational complexity.

[Distance 1] Using Slerp Interpolation

- **Spherical geodesic as path** : Taking spherical geodesic as the path between x and y
- **Field Assumption**: The drift vector field w changes linearly along this path

4

$$\omega(t) \approx (1-t)x_\omega + ty_\omega$$
$$\langle (1-t)x_\omega + ty_\omega, A(t)x_{id} + B(t)y_{id}\rangle$$
$$(1-t)A(t)\langle x_\omega, x_{id}\rangle + (1-t)B(t)\langle x_\omega, y_{id}\rangle + tA(t)\langle y_\omega, x_{id}\rangle + tB(t)\langle y_\omega, y_{id}\rangle$$

5

$$\mathcal{A}(\mathbf{x},\mathbf{y}) = \int_0^1 \langle \omega(\gamma(t)), \dot{\gamma}(t)\rangle dt \approx \sum_{k=1}^{K} \langle \omega(t_k), \dot{\gamma}(t_k)\rangle \Delta t$$
$$\Delta t = \frac{1}{K} \qquad t_k \in \{\frac{\Delta t}{2}, \frac{3\Delta t}{2}, ..., 1 - \frac{\Delta t}{2}\} \qquad K = 10$$

Critique: By defining the intermediate drift vector as such, we are linearly interpolating tangent vectors in the ambient Euclidean space. Mathematically inconsistent without applying Levi-Civita parallel transport

Employs a midpoint Riemann sum to discretely integrate the continuous path over K=10 steps

When numerically approximating the continuous integral over the interval $t \in [0, 1]$,the interval is discretized into K sub-intervals (bins), each with a width of $\Delta t = \frac{1}{K}$.

- Evaluating the continuous function at the exact center (midpoint) of each bin provides a significantly more accurate approximation than evaluating it at the left or right edges.

- **Bin 1:** Range $[0, \Delta t]$ .The midpoint is $0 + \frac{\Delta t}{2} = \frac{\Delta t}{2}$.
- **Bin 2:** Range $[\Delta, t2\Delta t]$ .The midpoint is $\Delta t + \frac{\Delta t}{2} = \frac{3\Delta t}{2}$.
- **Bin 3:** Range $[2, \Delta t3\Delta t]$ .The midpoint is $2\Delta t + \frac{\Delta t}{2} = \frac{5\Delta t}{2}$.
- ...
- **Bin K:** Range $[1, -\Delta t1]$ .The midpoint is $(1 - \Delta t) + \frac{\Delta t}{2} = 1 - \frac{\Delta t}{2}$.

By using the midpoint rule, the local error of the estimation is bounded by $O(\Delta t^2)$,effectively canceling out first-order linear errors compared to

standard left-endpoint or right-endpoint evaluations.

Critique:

- Tangent spaces at different points on a Riemannian manifold are disjoint vector spaces $T_x S^{n-1} \neq T_y S^{n-1}$
- You cannot linearly mix vectors from distinct tangent spaces without applying Levi-Civita parallel transport along the geodesic connecting them [Absil et al., Optimization on Matrix Manifolds 2008].

[Distance 1] Analytically solving the interpolation

Problem: Interpolate tangent vectors $x_\omega \in T_x\mathbb{S}^{n-1}$ and $y_\omega \in T_y\mathbb{S}^{n-1}$ along $\gamma(t)$

1. $T_{\gamma(t)}\mathbb{S}^{n-1}$, $\omega(t) = (1-t)P_{x\to\gamma(t)}(x_\omega) + tP_{y\to\gamma(t)}(y_\omega)$

2. $$\langle P(U), P(V)\rangle = \langle U, V\rangle$$

   SLERP velocity $\dot\gamma(t)$ is the tangent vector field of $\gamma(t)$ -> it is parallel to itself $\nabla_{\dot\gamma}\dot\gamma = 0$

   $\dot\gamma(t)$ Is the parallel transport of $\dot\gamma(0)$

3. $$\langle P_{x\to\gamma(t)}(x_\omega), \dot\gamma(t)\rangle \equiv \langle x_\omega, \dot\gamma(0)\rangle \qquad \langle P_{y\to\gamma(t)}(y_\omega), \dot\gamma(t)\rangle \equiv \langle y_\omega, \dot\gamma(1)\rangle$$

$$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \int_0^1 \left[(1-t)\langle x_\omega, \dot\gamma(0)\rangle + t\langle y_\omega, \dot\gamma(1)\rangle\right] dt = \frac{1}{2}\langle x_\omega, \dot\gamma(0)\rangle + \frac{1}{2}\langle y_\omega, \dot\gamma(1)\rangle$$

1. To linearly interpolate tangent vectors along the geodesic gamma, we need to parallel transport both vectors to the intermediate tangent space, where $P_{A \to B}$ denotes the Levi-Civita parallel transport operator.
2. By the metric compatibility of the Levi-Civita connection, parallel transport preserves inner products
3. Because the SLERP velocity $\dot\gamma(t)$ is the tangent vector field of an affinely parameterized geodesic, it is parallel to itself $\nabla_{\dot\gamma}\dot\gamma = 0$ .Consequently, $\dot\gamma(t)$ is exactly the parallel transport of its initial velocity $\dot\gamma(0)$
4. The inner product with the transported drift vector simplifies to its boundary projection, which simplifies the integral of the geodesic to a linear function
5. Given the exact boundary derivatives of SLERP,

[Distance 1] Analytically solving the interpolation

3 $$\langle P_{x\to\gamma(t)}(x_\omega), \dot\gamma(t)\rangle \equiv \langle x_\omega, \dot\gamma(0)\rangle \qquad \langle P_{y\to\gamma(t)}(y_\omega), \dot\gamma(t)\rangle \equiv \langle y_\omega, \dot\gamma(1)\rangle$$

$$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \int_0^1 \left[(1-t)\langle x_\omega, \dot\gamma(0)\rangle + t\langle y_\omega, \dot\gamma(1)\rangle\right] dt = \frac{1}{2}\langle x_\omega, \dot\gamma(0)\rangle + \frac{1}{2}\langle y_\omega, \dot\gamma(1)\rangle$$

4 $$\dot\gamma(0) = \frac{\theta}{\sin\theta}(y_{\text{id}} - x_{\text{id}}\cos\theta) \qquad \dot\gamma(1) = \frac{\theta}{\sin\theta}(y_{\text{id}}\cos\theta - x_{\text{id}})$$

$$\langle x_\omega, x_{\text{id}}\rangle = 0 \qquad \langle y_\omega, y_{\text{id}}\rangle = 0$$

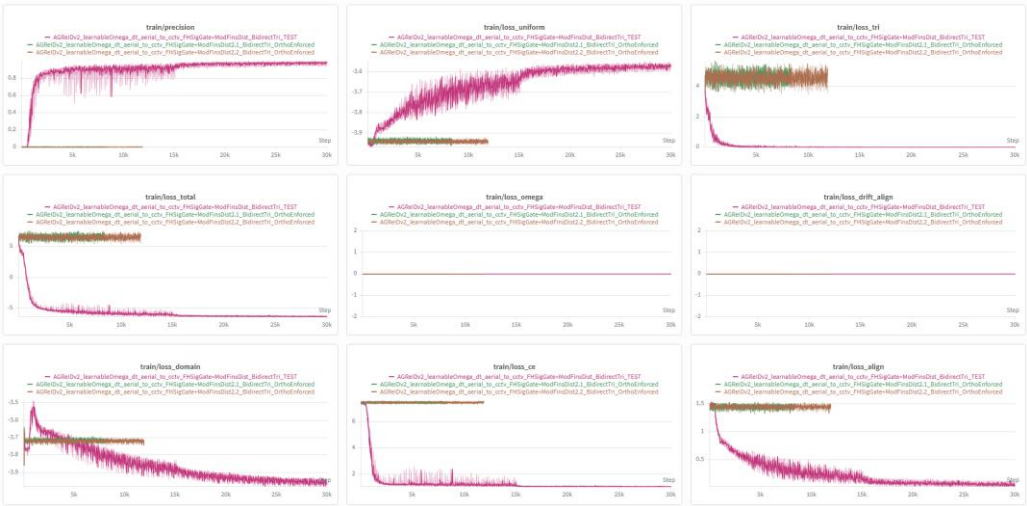
$$-\langle x_\omega, x_{\text{id}}\cos\theta\rangle = 0 \qquad \langle y_\omega, y_{\text{id}}\cos\theta\rangle = 0$$

5 $$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \frac{1}{2}\frac{\theta}{\sin\theta}\Big(\langle x_\omega, y_{\text{id}}\rangle - \langle y_\omega, x_{\text{id}}\rangle\Big)$$

Given the exact boundary derivatives of SLERP, we can assume strictly enforced tangent orthogonality to see that two terms evaluate to zero

The exact continuous integral resolves to

Results



- Loss curves are flatlining at exactly 0 and invariant total/triplet/CE losses indicate gradient explosion leading to network collapse

Hypothesis 1

The formulation introduces two well-documented numerical singularities in Riemannian backpropagation [Nickel & Kiela, NeurIPS 2017]:

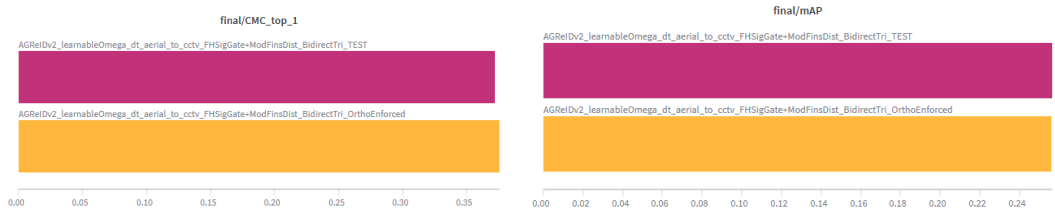
- **The Arccosine Singularity:** torch.acos(cos_theta) has a derivative of $-1/\sqrt{1-x^2}$
    - As cos_theta approaches 1, the gradient approaches infinity. Clamping to 1.0 - 1e-6 yields a maximum gradient multiplier of ~ 707
    - In a Triplet Loss formulation, this massive scalar propagates backward, causing an immediate weight explosion.
- **The Indeterminate Ratio:** The term theta / sin_theta evaluates to $0/0$ as $\theta \to 0$. Even with clamping sin_theta to 1e-6, if theta evaluates to 0.0, the gradient calculation inside PyTorch's auto-grad graph becomes highly unstable.

Detach the arccos?

[Model] Enforcing Tangent Plane Orthogonality

- For a point $x \in S^{n-1}$, the tangent space is $\{v \in R^n \mid \langle x, v \rangle = 0\}$
- **Apply** orthogonal projection operator of the hypersphere

The exact projection onto the tangent plane $T_x S^{n-1}$ is defined as $x_\omega = (I - x_{\mathrm{id}} x_{\mathrm{id}}^T)\hat{x_\omega}$



For a point $x \in S^{n-1}$, the tangent space is $\{v \in R^n \mid \langle x, v \rangle = 0\}$.To strictly enforce this, you must apply the orthogonal projection operator of the hypersphere[Absil et al., Optimization on Matrix Manifolds 2008].

Did not yield big results

[Regularization] Preventing Omega from Diverging

We use the logarithmic barrier penalty

$$s_i = \max(\epsilon, c - \|\omega_i\|_2)$$

Where c is the maximum norm defined (0.95).

$$\mathcal{L}_{\text{barrier}} = -\frac{1}{B}\sum_{i=1}^{B} \log(\max(\epsilon, c - \|\omega_i\|_2))$$

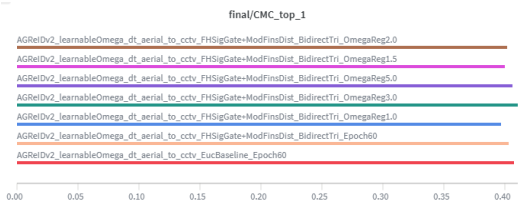
$$s_i \to 0 \qquad -\log(s_i) \to \infty$$

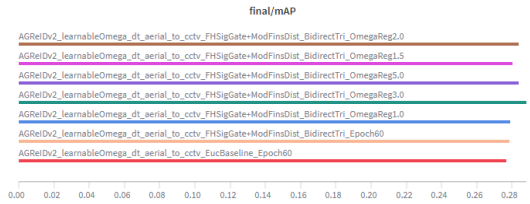
Gradients of this loss w.r.t. norm is:

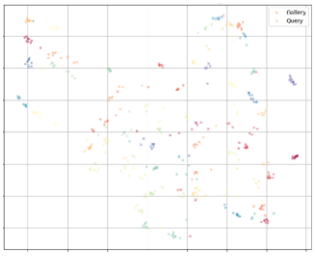
$$\frac{\partial \mathcal{L}_{\text{barrier}}}{\partial \|\omega_i\|_2} = \frac{1}{c - \|\omega_i\|_2}$$

- The slack variable s_i represents the distance from the norm to the boundary
    - To prevent numerical instability when evaluating the logarithm at exactly zero, a clamping parameter ϵ (eps = 1e-6) is introduced
- The final empirical loss across the batch is the mean of the negative logarithms of these slack variables
- From an optimization perspective, as the norm $\| \omega_i \|_2$ approaches the boundary c, the slack $s_i \to 0$,and the term $-\log(s_i) \to \infty$.
- This indicates that the restorative force repelling the vector from the boundary grows asymptotically to infinity as the vector approaches the boundary, acting as a strict "hard constraint."

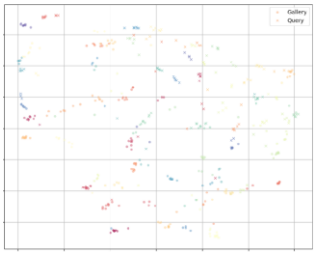
[Regularization] Preventing Omega from Diverging







OmegaReg=3.0



Baseline

[**Theory**]: Explain why drift align and bidirectional triplet are incompatible

- Applying both constraints simultaneously overconstrains the latent geometry
- Model does not have enough “slack” to model asymmetry?

Not quite sure why

[**Theory**]: Explain why having a 32-dimensional drift vector is optimal

Found that the premise of integrating an asymmetric Finsler space into a domain-generalizable person Re-ID space is theoretically flawed?

- Person Re-ID is fundamentally an identity retrieval task, which necessitates a equivalence relation governed by metric symmetry
- Introducing distance asymmetry inherently violates identity equivariance, as the retrieval ranking becomes strictly dependent on the query-gallery orientation.

Drift vector of size 32 is "optimal" and surpasses the Euclidean baseline:

- not indicative of the model successfully learning meaningful domain-based asymmetry
- an artifact of information bottlenecking and implicit feature-space regularization

1. Capacity Constraint on Asymmetry
2. Topological Perturbation vs. Uniformity Collapse

Capacity Constraint on Asymmetry

- asymmetry fundamentally degrades the symmetric identity matching objective
- slicing the drift vector to a restricted dimensional subspace (e.g., 32 dimensions) heavily limits the destructive impact of the asymmetric term in the Randers metric
- Bounding the drift to 32 dimensions forces the network to allocate the vast majority of the latent space capacity to the symmetric Riemannian (Euclidean) metric, effectively preserving the identity equivalence class.
- Increasing the drift dimension (e.g., 64, 128, 2048) allocates excessive capacity to the directional bias, catastrophically degrading symmetric retrieval performance as demonstrated in your mAP and Rank-1 drops.

Topological Perturbation vs. Uniformity Collapse:

- The BAU framework mitigates the polarized effects of data augmentations by explicitly balancing alignment and uniformity on an isotropic unit hypersphere.
- The Randers metric introduces a local preferred direction via the drift vector, which inherently breaks this required hyperspherical isotropy.
- A high-dimensional drift (e.g., ≥ 64) aggressively warps the manifold

    - Encouraging spatial sparsity and representation collapse that the BAU uniformity loss cannot recover from.
- A 32-dimensional bottleneck acts as an optimal regularizer
    - it provides enough geometric perturbation to prevent source-domain overfitting
    - while maintaining sufficient global isotropy for the uniformity loss to converge.
    - A 16-dimensional drift simply lacks the parameterization to exert meaningful regularization gradients over the Euclidean baseline.

Next Week

1. **[Model] Manifold Smoothness Regularization:** Force the network's vector field to match the surrogate.
   1. During training, sample the midpoint of the geodesic $\gamma(0.5)$
   2. Pass this midpoint through the neural network to obtain the true intermediate drift vector ω_{true}
   3. Compute the parallel-transported surrogate vector at the midpoint, $\omega_{surrogate}$.
   4. Add a regularization loss: $L_{smooth} = \| \omega_{true} - \omega_{surrogate} \|_2^2$

2. **[Model] Domain-Conditioned Global Drift:** Define learnable, global drift vectors anchored to specific domains (datasets in BAU).
   1. A constant vector field defined by parallel transport is inherently exact, bypassing the surrogate field error entirely.

3. **[Visualisation] Plot the omega drift vector**
4. Take the 2025 paper and migrate the Finsler modifications and see where we are at
   - Still want me to the academic work
   - Still want me to perform

By using the analytical parallel-transport solution, you are assuming that the intermediate neural vector field can be perfectly modeled by linearly interpolating the boundary vectors $\omega(x)$ and $\omega(y)$.

- This interpolated field is a "surrogate." Because neural networks are highly non-linear, the actual drift vector the network *would* predict halfway along the path, $\omega(\gamma(0.5))$,will likely diverge entirely from the linearly interpolated approximation. You are imposing a geometric prior (smooth, parallel-transported flow) onto a network that has no incentive to learn a flow that obeys this prior.

Manifold Smoothness Regularization: This coerces the neural field into a state where your analytical approximation is geometrically valid

Domain-Conditioned Global Drift:

- Instead of predicting identity-specific drift vectors (which creates a chaotic, non-linear vector field)